BP-R Stage 3: source laws and merge choices

6 October 2026 — supplied originals read; S3-G2-N open

Scope. Source transcriptions and static proposed-node evaluations. Neither node is adopted. No simulator, traffic experiment or gradient validation.

1. What the supplied journal actually specifies

Laval and Daganzo (2006), Section 2.1, equations 6–13, specifies continuous lane traffic with a triangular fundamental diagram. In this unit-consistent transcription, k_ℓ is donor density (veh/m), u free-flow speed (m/s), $w > 0$ backward wave-speed magnitude (m/s), κ jam density (veh/m), Q lane capacity (veh/s), and Δt a time interval (s). Sending mass:

$$S_\ell = \Delta t \min\{uk_\ell, Q\}. \tag{1}$$

For allowed target lane b, lane-choice speeds $\bar{v}$ (m/s) and relaxation time τ (s), desired lane-change rate and transfer masses are

$$\pi_{\ell b} = \frac{(\bar{v}_b - \bar{v}_\ell)_+}{u\tau}, \qquad d_{\ell b} = \pi_{\ell b}\Delta t\, S_\ell, \quad b \neq \ell, \tag{2}$$

$$d_{\ell\ell} = \left(1 - \sum_{b\neq\ell} \pi_{\ell b}\Delta t\right) S_\ell. \tag{3}$$

$(z)_+ = \max(0, z)$. The step must give $\sum_{b\neq\ell} \pi_{\ell b}\Delta t \leq 1$; outgoing demands sum to S_ℓ. Source lane changes advance from cell (i, ℓ) to $(i + 1, b)$; they are not counted twice as separate lateral/longitudinal moves.

For target cell/lane b, collect all compatible incoming demands d_{jb} and form receiving mass and actual transfers:

$$R_b = \Delta t \min\{w(\kappa - k_b), Q\}, \qquad \gamma_b = \min\left\{1, \frac{R_b}{\sum_j d_{jb}}\right\}, \qquad f_{jb} = \gamma_b d_{jb}. \tag{4}$$

If demand is zero, all transfers are zero; no ratio is evaluated. This transcribes the journal's local IT allocation in mass units. Transfers respect receiving and desired donor demand. Pair each debit and credit:

$$n_c^+ = n_c + \sum_j f_{jc} - \sum_b f_{cb}. \tag{5}$$

Boundary admissions/exits are separate. Internal transfers cancel in the sum; event counts never round n_c.

Speed and receiving scope. Lane-choice speed uses the journal's downstream look-ahead/preceding-time space-mean convention, not arbitrary point FD inversion. Equation (4) is the unobstructed local recipe. Active moving constraints must use the coupled source method; do not reuse unobstructed supply and add a guessed acceleration penalty.

2. The motion law is now source-backed

Appendix A, equations A.1–A.4, uses disturbance position $\phi(t)$ (m), realized speed $v(t)$ (m/s), desired kinetic speed $v_{\text{des}}(t)$, and immediately downstream KW-stream speed $v_{\text{dwn}}(t)$:

$$v(t) = \min\{v_{\text{dwn}}(t), v_{\text{des}}(t)\}, \tag{6}$$

$$v_{\text{des}}(t + \Delta t) = v(t) + a(v(t), \phi(t))\Delta t, \tag{7}$$

$$\phi(t + \Delta t) = \phi(t) + v(t)\Delta t, \tag{8}$$

$$a(v, x) = a_0(1 - v/v_{\max}) - gG(x). \tag{9}$$

a_0 is zero-speed acceleration (m/s^2), $v_{\max}$ vehicle free maximum speed (m/s), g gravity (m/s^2), and $G(x)$ dimensionless grade. Traffic constrains realized speed; $v_{\max}$ need not equal u. Numerical examples in the journal and merge manuscript have different parameters; none is silently combined or calibrated.

v_{dwn} is a numerical estimate over the ensuing interval on the occupied lane, computed from the KW problem affected by the constraint. It is not independently assigned from the upstream command. Zero passing *relative to the moving disturbance* permits positive stationary-detector flow.

For cumulative actual transfer mass $H(t)$, the deterministic option creates descriptors when $\lfloor H(t) \rfloor$ increases. Only descriptors are discrete; the conserved transfer in (5) remains continuous, including fractional mass. Traffic is already counted in the field. Descriptors are not extra TST population. A complete task must declare birth speed, event phase and insertion timing. No random sampling or coincidence jitter is approved.

Numerical lineage. Section 2.2 cites the 2005 method that converges in flows; Appendix A cites the separate numerical-treatment paper. Available 2003 originals distinguish cumulative-count approximation from continuous-space composite-Riemann coupling. The latter uses wave-interaction and crossing step conditions. This supports a conservative route, not a theorem for every endogenous event/control derivative.

Second supplied paper. Daganzo, Lin and Del Castillo (1997), Sections 2–3, proves IT equivalence for two classes with different permissions to two lane sets. Its priority vehicles are an access class. Section 4 illustrates an exit-ramp/diverge; Section 5 leaves general junction extension open. It does not uniquely determine the merge manuscript's 40% adjustment.

Preserved task. Control caps admission to the finite KW ramp approach. Merge entry follows later demand and receiving; it is not also capped by the contemporaneous command. All waiting, ramp and mainline inventory is in TST. There is no added recovery-cost term.

3. Remaining choice: mandatory point merge

$D_r, D_m \geq 0$ are ramp and non-ramp incoming demand rates (veh/s); $R \geq 0$ is shared target receiving (veh/s), including existing moving-constraint restrictions. Non-ramp demand includes all compatible through/lateral inflows. No other movement may spend R again. Actual rates are f_r, f_m.

A: fixed junction priority. A fixed ramp share $p_r \in (0, 1)$ is common to all controls/models. Daganzo's 1994 original, Section 3.2, supplies the two-approach construction, transcribed as

$$(f_r, f_m) = \begin{cases} (D_r, D_m), & D_r + D_m \leq R, \\ (\min\{D_r, \max(p_r R, R - D_m)\},\ R - f_r), & D_r + D_m > R. \end{cases} \tag{10}$$

Unused shares are reallocated. Grouping local lane demands and applying this node to the selected reference is a proposed adaptation, not exact reproduction of the unread gridlock chapter. No p_r value is inferred from 40%.

B: demand-proportional allocation. Extend local IT to the mandatory node:

$$(f_r, f_m) = \begin{cases} (D_r, D_m), & D_r + D_m \leq R, \\ \dfrac{R}{D_r + D_m}(D_r, D_m), & D_r + D_m > R. \end{cases} \tag{11}$$

No fixed share is introduced. Contested fractions change with demands. This extension is not a general-junction theorem from the 1997 paper.

Both give $0 \leq f_j \leq D_j$ and $f_r + f_m = \min(D_r + D_m, R)$. In (10)'s overloaded case, $R - D_m \leq f_r \leq D_r$ when the lower bound is positive, and $0 \leq f_r \leq R$; hence $0 \leq R - f_r \leq D_m$. In (11), the common multiplier is between zero and one. These are static algebraic checks, not traffic runs.

R	D_m	D_r	A: f_m, f_r	B: f_m, f_r
100	90	90	70, 30	50, 50
100	20	90	20, 80	18.18, 81.82
100	90	10	90, 10	90, 10
100	0	90	0, 90	0, 90
0	90	90	0, 0	0, 0

Illustrative consistent rate units; A uses $p_r = 0.30$ only for these examples. Proportional allocation can leave some mainline demand unserved although it is below A's fixed share. Both use all available receiving.

Recommendation and limits. A has direct merge-specific primary support. B avoids a share parameter but assumes demand-based competition. Either needs adoption at S3-G2-N. Then qualify joined event/receiving timing, clearance, inactive equivalence, strong independent closures and actual rate leverage. No full-TST benefit, fine-state necessity, useful gradient or Stage 3 closure is established here.